

Retail Sales Forecasting & Demand Analytics Internship Project Report

Retail Sales Analysis • Demand Forecasting • Inventory Intelligence
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1. Executive Summary

This report presents the findings of a data-driven analysis of our retail sales operations, conducted without the need for any specialist technical knowledge to interpret. Over the review period, overall sales demonstrated a consistent upward growth trend, with predictable seasonal fluctuations concentrated in specific calendar months. Advanced forecasting tools were applied to project future demand with measurable reliability. Unusual sales events — both spikes and unexpected dips — were automatically detected and explained. Finally, our product catalogue was segmented into three distinct demand groups, enabling more targeted inventory planning. Together, these insights provide a clear, actionable foundation for improving stock management, demand planning, and supply chain efficiency.

2. Key Findings

Steady Revenue Growth:	Sales have grown consistently over the full analysis period, confirming a healthy and expanding demand base.
Seasonal Demand Peaks:	Recurring demand spikes occur during specific months, providing a reliable planning calendar for procurement teams.
Best Forecast Model:	Among three tested models (SARIMA, Prophet, XGBoost), Prophet delivered the highest forecasting accuracy and is recommended for ongoing use.
High-Revenue Products:	Phones and Chairs consistently generate the highest sales volumes, making them the highest-priority items for stock reliability.
Unstable Low-Demand Products:	A subset of lower-selling products showed irregular purchasing patterns, signalling over-stocking risk.

3. 3-Month Sales Forecast

Based on historical patterns, sales are expected to continue their gradual upward trajectory over the next three months. Seasonal fluctuations are likely to persist and should be anticipated during known peak periods. The forecast model provides confidence intervals, meaning actual sales will most likely fall within a defined upper and lower band — reflecting moderate prediction reliability. Planning inventory around the midpoint of these ranges, while keeping buffer stock for the upper bound, is the recommended approach.

4. Top Anomalies Detected

Anomaly 1 — Sales Spike in High-Demand Categories	A sudden and significant surge in sales was detected in top-performing product categories. This is most likely attributable to a promotional campaign, seasonal offer, or bulk purchase event.
Anomaly 2 — Unusually Low Sales in Specific Weeks	Certain weekly periods recorded sales well below the normal baseline. Probable causes include stock shortages, supply chain delays, or reduced consumer activity during off-peak periods.
Anomaly 3 — Irregular Demand in Low-Performing Products	Erratic purchasing behaviour was observed in the lower-demand product segment. This may reflect inconsistent promotional support or shifting consumer preferences in those categories.

5. Product Demand Segmentation & Stocking Strategy

Segment	Products	Demand Profile	Recommended Strategy
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Cluster 2 (High)	Phones, Chairs	High, consistent volume	Maintain elevated safety stock; prioritise in replenishment cycles
Cluster 3 (Medium)	Mid-range products	Stable, moderate demand	Standard stock levels; target for growth marketing
Cluster 1 (Low)	Niche / slow-movers	Limited, irregular sales	Reduce holdings; avoid over-procurement

6. Business Recommendations

Recommendation 1: Prioritise Stock for High-Demand Products

Phones and Chairs consistently account for the highest sales volumes. Maintaining higher safety stock for these items will reduce the risk of stockouts during peak demand periods and protect revenue continuity.

Recommendation 2: Embed Forecasting into Supply Chain Planning

The Prophet model demonstrated reliable accuracy in predicting future demand. Integrating its 3-month rolling forecast into the procurement cycle will enable more precise purchase orders and reduce both overstocking and shortage events.

Recommendation 3: Act on Anomaly Alerts Proactively

Regular monitoring of sales anomalies allows the business to respond quickly to unexpected demand shifts. Early identification of spikes enables rapid restocking, while early detection of dips can trigger timely promotional or clearance actions.

7. Risk & Limitation

Important Notice for Decision-Makers: The forecasting system is built on historical sales data and assumes that future market conditions will broadly follow past patterns. Should unexpected external factors arise — such as significant economic changes, supply disruptions, or major shifts in consumer behaviour — forecast accuracy may decline. It is recommended that forecast outputs are reviewed alongside current market intelligence and adjusted by business judgment where necessary. This system is a decision-support tool, not a replacement for informed human oversight.